HSR-ACC-27 Access Node Hardware Environmental Fault Resolution SOP for Arista CLI Dialect

Node: HSR-ACC-27 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_1100

Description

This SOP outlines the procedures for identifying and resolving a hardware/environmental fault on the HSR-ACC-27 access node, specifically within the Arista CLI dialect. The failure domain is primarily attributed to ASIC memory parity errors, line card crashes, or power supply redundancy loss. The root cause of this issue is often linked to the intrinsic hardware design of the Arista EOS (Extensible Operating System) and its reliance on the ASIC (Application-Specific Integrated Circuit) for processing and forwarding network traffic. The ASIC's memory parity error detection and correction mechanisms can sometimes fail, leading to line card crashes or power supply redundancy loss. Furthermore, environmental factors such as temperature fluctuations, power supply issues, or physical damage to the node can also contribute to this failure domain. In this SOP, we will delve into the diagnostic and resolution procedures for addressing this critical issue, ensuring minimal downtime and optimal network performance.

Pre-requisite Checks

- Check: Confirm the presence of redundant links and power supplies at the HSR Layout Access site, and verify their operational status.

- Check: Dispatch remote hands to the HSR Layout Access site to visually inspect the node and surrounding environment for any signs of physical damage or environmental anomalies.

- Check: Verify the node's temperature and humidity levels, ensuring they are within the recommended operating range.

Potential Root Causes

Cause	Probability
ASIC memory parity errors	High
Line card crashes	Medium
Power supply redundancy loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show system resource, Expected Output: System Resource Utilization: ASIC Memory: 75% Line Card Status: Online Power Supply Status: Redundant, Description: This command isolates the current system resource utilization, allowing us to assess the ASIC memory usage and line card status, thereby pinpointing potential issues.
```

```
Command: show environment temperature, Expected Output: Temperature: Current Temperature: 25°C Threshold: 40°C, Description: This command provides the current temperature and threshold values, enabling us to determine if the temperature is within the recommended operating range.

Command: show power supply status, Expected Output: Power Supply Status: Redundant: Online Backup: Offline, Description: This command verifies the power supply status, ensuring redundant power is available and functional.
```

Resolution Steps

Step 1: Step: Replace the faulty ASIC or line card, Command: replace system resource asic, Description: This step involves replacing the faulty ASIC or line card with a new one, ensuring optimal network performance and minimizing downtime.

Step 2: Step: Restore power supply redundancy, Command: power supply status restore, Description: This step involves restoring the power supply redundancy, ensuring uninterrupted network operation.

Step 3: Step: Reset the node, Command: system reset, Description: This step involves resetting the node to its default configuration, ensuring a clean slate for further troubleshooting and resolution.

Post-Check Verification

```
Command: show system resource, Expected Output: System Resource Utilization: ASIC Memory: 50% Line Card Status: Online Power Supply Status: Redundant, Description: This command verifies the system resource utilization, ensuring the ASIC memory usage and line card status are within optimal ranges.

Command: show environment temperature, Expected Output: Temperature: Current Temperature: 25°C Threshold: 40°C, Description: This command verifies the temperature, ensuring it is within the recommended operating range.

Command: show power supply status, Expected Output: Power Supply Status: Redundant: Online Backup: Online, Description: This command verifies the power supply status, ensuring redundancy is restored and functional.
```

Roll-back Procedures

- Step: Restore the original configuration, Command: system restore, Description: This step involves restoring the original configuration, ensuring the node returns to its previous state.

- Step: Revert the changes, Command: revert changes, Description: This step involves reverting the changes made during the resolution process, ensuring the node is restored to its original state.